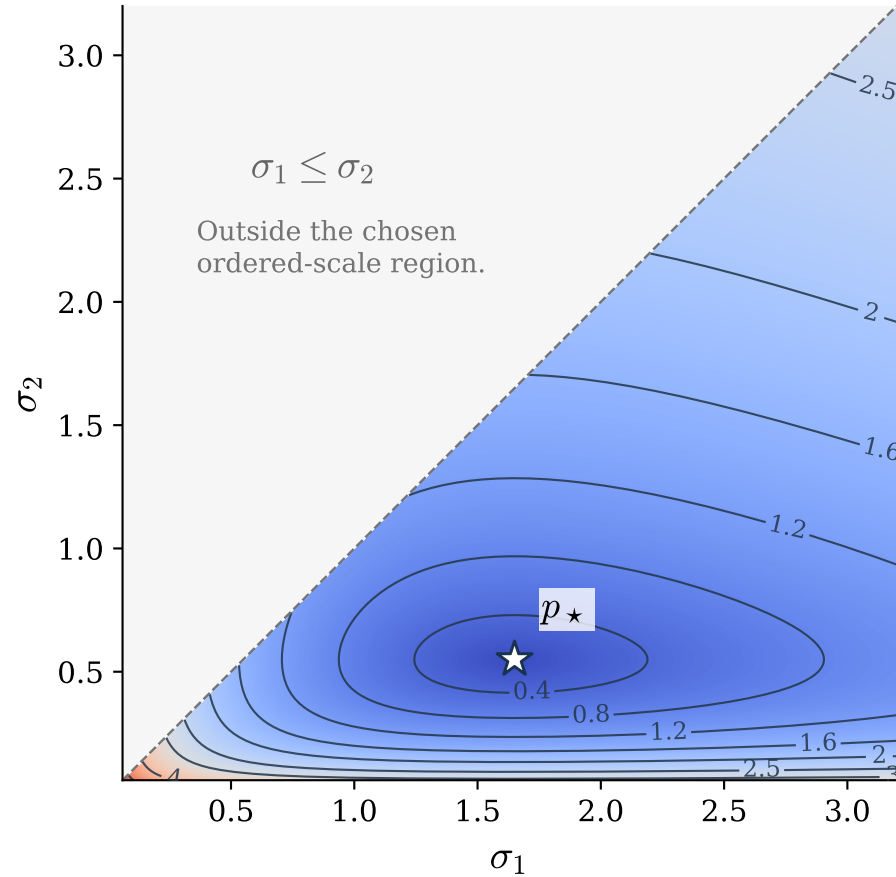


# Gaussian distance maps in principal-scale coordinates

$$\mu = (0.5, -0.5)^T, \quad \alpha = 30^\circ \quad (\text{fixed}), \quad \Sigma = R(\alpha) \text{diag}(\sigma_1^2, \sigma_2^2) R(\alpha)^T$$

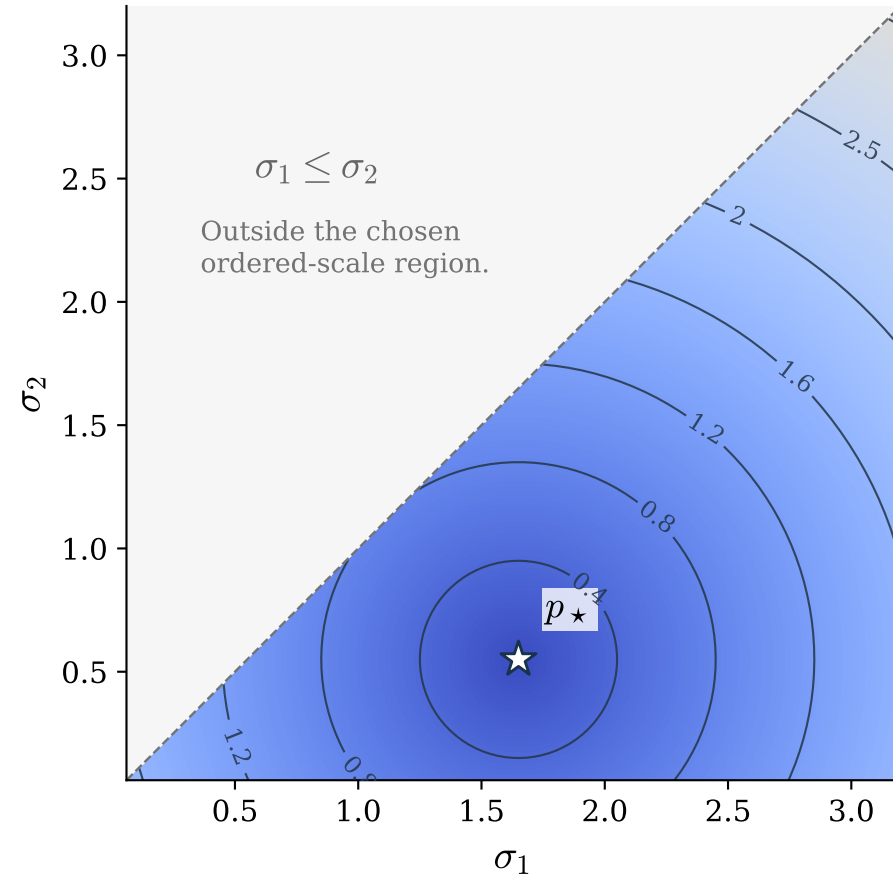
Reference:  $(\sigma_1, \star, \sigma_2, \star) = (1.65, 0.55)$

Fisher-Rao

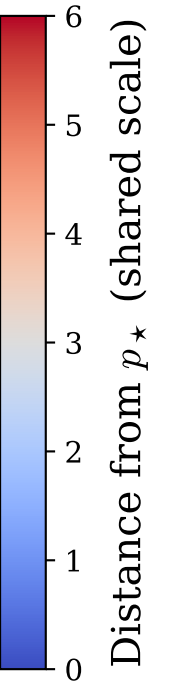


$$d_{\text{FR}} = \sqrt{2 \left[ \log^2 \left( \frac{\sigma_1}{1.65} \right) + \log^2 \left( \frac{\sigma_2}{0.55} \right) \right]}$$

Wasserstein-Otto



$$d_{\text{WO}} = \sqrt{(\sigma_1 - 1.65)^2 + (\sigma_2 - 0.55)^2}$$



Fisher-Rao measures relative scale changes; Wasserstein-Otto measures absolute scale changes.

Identical axes, contour levels, and color scale. The diagonal marks equal principal scales and is excluded.